ECEN-601: Homework 1

Due on September 12, 2025 at 5:00 pm

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Problem 1

Which of the following expressions are statements?

(a) Today is a nice day.

(b) Go to sleep.

(c) Is it going to snow tomorrow?

(d) The U.S. has 49 states.

(e) I like to eat fruit, and you often think about traveling to Spain.

(f) If we go out tonight, the babysitter will be unhappy.

(g) Call me on Thursday if you are home.

Problem 2

Let $X =$ "I am happy," let $Y =$ "I am watching a movie" and $Z =$ "I am eating spaghetti." Translate the following statements into words.

(a) $Z \to X$.

(b) $X \leftrightarrow Y$.

(c) $(Y \vee Z) \to X$.

(d) $Y \vee (Z \to X)$.

(e) $(Y \to \neg X) \wedge (Z \to \neg X)$.

(f) $(X \wedge \neg Y) \leftrightarrow (Y \vee Z)$.

Solution

Problem 3

Let E = "The house is blue," let F = "The house is 30 years old" and G = "The house is ugly." Translate the following statements into symbols.

(a) If the house is 30 years old, then it is ugly.

(b) If the house is blue, then it is ugly or it is 30 years old.

(c) If the house is blue, then it is ugly, or it is 30 years old.

(d) The house is not ugly if and only if it is 30 years old.

(e) The house is 30 years old if it is blue, and it is not ugly if it is 30 years old.

(f) For the house to be ugly, it is necessary and sufficient that it be ugly and 30 years old.

Solution

Problem 4

Make a truth table for each of the following statements.

(a) $P \wedge \neg Q$.

(b) $(R \vee S) \wedge \neg R$.

(c) $X \vee (\neg Y \vee Z)$.

(d) $(A \vee B) \wedge (A \vee C)$.

(e) $(P \wedge R) \vee \neg(Q \wedge S)$.

Solution

Problem 5

Simplify the following statements (making use of any equivalance of statements given so far in the text or exercises).

(a) $\neg(P \to \neg Q)$

(b) $A \to (B \wedge A)$

(c) $(X \wedge Y) \to X$

(d) $\neg(M \vee L) \wedge L$

(e) $(P \to Q) \vee Q$

(f) $\neg(X \to Y) \vee Y$

Solution

Problem 6

Suppose that the possible values of x are all people. Let $Y(x)$ = "x has green hair," let $Z(x)$ = "x likes pickles" and $W(x)$ = "x has a pet frog." Translate the following statements into words.

(a) $(\forall x)Y(x)$.

(b) $(\exists x)Z(x)$.

(c) $(\forall x)[W(x) \wedge Z(x)]$.

(d) $(\exists x)[Y(x) \rightarrow W(x)]$.

(e) $(\forall x)[W(x) \leftrightarrow \neg Z(x)]$.

Solution